```
import numpy as np
from scipy.linalg import lu,lu_factor, lu_solve
A=np.array(([3,-1,2],[-3,-2,10],[9,-5,6]), dtype=float)
b=np.array([10,20,30],dtype=float)
P,L,U=lu(A)
print("Matrix A: \n",A)
print("Matrix Permutation: \n",P)
print("Matrix Lower: \n",L)
print("Matrix Upper: \n",U)
lu_factorization = lu_factor(A)
print("LU Factorization: \n",lu_factorization)
x=lu_solve(lu_factorization,b)
print("Solution: \n",x)
```

```
Matrix A:
 [[ 3. -1.  2.]
 [-3. -2. 10.]
 [ 9. -5.  6.]]
Matrix Permutation:
 [[0. 0. 1.]
 [0. 1. 0.]
 [1. 0. 0.]]
Matrix Lower:
 [[ 1.          0.          0.        ]
 [-0.33333333  1.          0.        ]
 [ 0.33333333 -0.18181818  1.        ]]
Matrix Upper:
 [[ 9.         -5.          6.        ]
 [ 0.         -3.66666667 12.        ]
 [ 0.          0.          2.18181818]]
LU Factorization:
 (array([[ 9.        , -5.        ,  6.        ],
       [-0.33333333, -3.66666667, 12.        ],
       [ 0.33333333, -0.18181818,  2.18181818]]), array([2, 1, 2], dtype=int32))
Solution:
 [ 1.66666667 -0.          2.5       ]
```

```
#
import numpy as np
from scipy.linalg import lu,lu_factor, lu_solve
A=np.array(([3,-1],[4,1]), dtype=float)
b=np.array([1,6],dtype=float)
P,L,U=lu(A)
print("Matrix A: \n",A)
print("Matrix Permutation: \n",P)
print("Matrix Lower: \n",L)
print("Matrix Upper: \n",U)
lu_factorization = lu_factor(A)
print("LU Factorization: \n",lu_factorization)
x=lu_solve(lu_factorization,b)
print("Solution: \n",x)
```

```
Matrix A:
 [[ 3. -1.]
 [ 4.  1.]]
Matrix Permutation:
 [[0. 1.]
 [1. 0.]]
Matrix Lower:
 [[1.   0.  ]
 [0.75 1.  ]]
Matrix Upper:
 [[ 4.    1.  ]
 [ 0.   -1.75]]
LU Factorization:
 (array([[ 4.  ,  1.  ],
       [ 0.75, -1.75]]), array([1, 1], dtype=int32))
Solution:
 [1. 2.]
```

```
#user input equation
a1,b1,c1,d1=map(float,input("Enter the equation 1 (a1,b1,c1,d1):").split())
a2,b2,c2,d2=map(float,input("Enter the equation 2 (a2,b2,c2,d2):").split())
a3,b3,c3,d3=map(float,input("Enter the equation 3 (a3,b3,c3,d3):").split())
A=np.array(([a1,b1,c1],[a2,b2,c2],[a3,b3,c3]), dtype=float)
b=np.array([d1,d2,d3],dtype=float)
P,L,U=lu(A)
print("Matrix A: \n",A)
print("Matrix Permutation: \n",P)
print("Matrix Lower: \n",L)
print("Matrix Upper: \n",U)
lu_factorization = lu_factor(A)
print("LU Factorization: \n",lu_factorization)
```

```
x=lu_solve(lu_factorization,b)
print("Solution: \n",x)
```

```
Enter the equation 1 (a1,b1,c1,d1):2 -4 4 -2
Enter the equation 2 (a2,b2,c2,d2):6 -9 7 -3
Enter the equation 3 (a3,b3,c3,d3):1 -4 8 0
Matrix A:
 [[ 2. -4.  4.]
 [ 6. -9.  7.]
 [ 1. -4.  8.]]
Matrix Permutation:
 [[0. 0. 1.]
 [1. 0. 0.]
 [0. 1. 0.]]
Matrix Lower:
 [[1.         0.         0.        ]
 [0.16666667 1.         0.        ]
 [0.33333333 0.4        1.        ]]
Matrix Upper:
 [[ 6.         -9.          7.        ]
 [ 0.         -2.5         6.83333333]
 [ 0.          0.         -1.06666667]]
LU Factorization:
 (array([[ 6.        , -9.        ,  7.        ],
       [ 0.16666667, -2.5       ,  6.83333333],
       [ 0.33333333,  0.4       , -1.06666667]]), array([1, 2, 2], dtype=int32))
Solution:
 [2.5   2.875 1.125]
```

```
import pandas as pd

# Load the file from Colab's default storage path
df= pd.read_csv('/content/book_sale.csv')

# View the first 5 rows
print(df.head())

A=df.iloc[:,0:3].to_numpy()
b=df.iloc[:,3].to_numpy()
print('matrix A\n',A)
print('vector b\n',b)
```

```
   popularity  stock  units sold  total sales
0          25     25           8           25
1          12     35          20           12
2          33     15           7           23
matrix A
 [[25 25  8]
 [12 35 20]
 [33 15  7]]
vector b
 [25 12 23]
```